FINAL PROJECT REPORT

Global Olympic Sports Performance Analysis

A Tableau-based visual analysis of Olympic medal performance, athlete participation, regional patterns, host-country performance, and investment categories.

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This report follows the supplied project-report format and distinguishes implemented work from potential future improvements.

1. INTRODUCTION

1.1 Project Overview

Global Olympic Sports Performance Analysis is a data-visualization project that uses institutional CSV datasets and Tableau to explore Olympic medal performance and athlete participation. The workbook brings together country rankings, medal-type comparisons, year-wise medal trends, athlete gender and sport participation, regional comparisons, host-country performance, and investment-category comparisons.

The project uses four CSV files: *alltime_country_rankings.csv*, *athletes.csv*, *games_summary.csv*, and *medal_results.csv*. The files are connected directly to Tableau, with relationships configured to support worksheets and dashboards.

1.2 Purpose

The purpose is to make Olympic performance data easier to explore through interactive dashboards and a narrative Story. The visualizations help users compare countries, review medal patterns over time, and explore athlete and regional information.

Project Objectives

- Compare countries by total Olympic medals and medal type.
- Explore changes in medal counts across Olympic years.
- Examine athlete participation by gender and sport.
- Compare regional medal efficiency and average medal counts across investment categories.
- Present host-country performance and key findings through Tableau dashboards and a four-point Story.

2. IDEATION PHASE

2.1 Problem Statement

Olympic datasets contain information across countries, athletes, games, sports, and medal results. When stored in separate files, it can be difficult to compare performance and identify patterns quickly. This project organizes the available data in Tableau and presents relevant comparisons through interactive visualizations.

2.2 Empathy Map Canvas

Target user: a sports data analyst, researcher, student, or sports enthusiast who wants to understand Olympic performance through clear visual evidence.

- **Thinks and feels:** Wants reliable comparisons and clear evidence behind Olympic performance.
- **Sees:** Large datasets with different fields and multiple dimensions of Olympic results.
- **Says and does:** Compares countries, years, sports, and medal categories to answer analytical questions.
- **Pains:** Manual comparison takes time; separate datasets can be difficult to interpret together.
- **Gains:** Faster exploration, clearer trends, and an organized overview of the data.

2.3 Brainstorming

- Use country rankings to compare total medal counts.
- Compare gold, silver, and bronze medals by country.
- Use a year-wise trend chart to explore changes over time.
- Analyze athlete participation by gender and sport.
- Compare regional efficiency, investment categories, and host-country performance.
- Combine dashboards into a four-point Story for a guided presentation.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User Action	Dashboard Support
Discover	Opens the project and identifies the main views.	Dashboard titles and organized charts.
Explore	Selects Year, Season, or Sport where available.	Interactive filters on configured dashboards.
Compare	Compares countries, medal types, sports, and regions.	Rankings, comparisons, and trend visualizations.
Understand	Reviews patterns and differences.	Dashboard arrangement and four-point Story.

3.2 Solution Requirement

- Connect and inspect the four supplied CSV datasets in Tableau.
- Configure relationships needed for cross-dataset analysis.
- Create visualizations for country, medal, year, athlete, regional, host, and investment comparisons.
- Provide Year, Season, and Sport filters where configured.
- Create three dashboards and a four-point Story.
- Validate that dashboards and configured filters behave as intended.

3.3 Data Flow Diagram

Data flow: Institutional CSV datasets → Tableau data connections and relationships → Field review and analysis → 8 worksheets / visualizations → 3 dashboards → 4-point Story → End user.

The project uses direct CSV connections in Tableau. It does not use a Snowflake warehouse or a separate SQL staging/curated/mart pipeline.

3.4 Technology Stack

Layer	Technology / Resource	Role
Data source	Four institutional CSV files	Stores country rankings, athlete, games, and medal data.
Data analysis	Tableau	Connects datasets, configures relationships, and builds visualizations.
Calculated field	Unique Athletes	Supports distinct athlete analysis; exact formula should match the workbook.
Presentation	Tableau dashboards and Story	Presents interactive comparisons and findings.
Repository	GitHub	Hosts project files and related documentation as available.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The solution fits the problem because it turns separate CSV files into related visual analytical views. Country rankings and medal comparisons answer performance questions, while athlete, regional, host-country, and investment-category views provide additional dimensions for exploration.

4.2 Proposed Solution

The solution is a Tableau workbook containing eight worksheets/visualizations, three dashboards, and a four-point Story. Interactive filters are available where configured. The project focuses on descriptive comparisons rather than predicting outcomes or claiming causal relationships.

4.3 Solution Architecture

Component	Description
Input	Four institutional CSV datasets.
Connection and preparation	Direct Tableau connections, field review, and configured dataset relationships.
Analysis and visualization	Eight worksheets covering medal, athlete, regional, host, and investment comparisons.
Reporting	Three dashboards and a four-point Tableau Story.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project can be organized into the following phases. This phase breakdown summarizes workstreams; it is not a claim that formal sprint tracking or time estimates were recorded for every activity.

Phase	Activities	Deliverable
Data setup	Connect CSVs and inspect field structure.	Connected Tableau data sources.
Data modeling	Configure relationships and review fields.	Data model ready for analysis.
Visualization	Build and refine eight worksheets.	Eight visualizations.
Dashboard and Story	Arrange worksheets, filters, and Story points.	Three dashboards and four-point Story.
Validation and reporting	Test configured filters and document the project.	Performance test evidence and final report.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Testing focused on checking that source data was visible in Tableau, relationships were configured, and worksheets, dashboards, and the Story rendered. Filters were tested by changing selections and reviewing whether relevant visualizations responded as configured.

Parameter	Evidence / Value
Data rendered	Four CSV datasets connected; row and field counts reviewed.
Data preprocessing	Field and data-type review; relationships configured in Tableau.
Filters	Year, Season, and Sport filters on configured dashboards.
Calculated field	Unique Athletes; verify formula against the workbook.
Dashboard design	8 worksheets / visualizations across 3 dashboards.
Story design	4 Story points.

Note: This report does not claim that SQL cleaning, automated duplicate removal, Snowflake processing, or a Flask application was completed.

7. RESULTS

7.1 Output Screenshots

Insert screenshots from the actual Tableau workbook in this section. Recommended evidence includes the Data Source page with the four connected datasets, the three dashboards, the calculated field editor for Unique Athletes, and the four-point Story. Use screenshots captured from the project rather than illustrative reference images.

Key Visualizations

- Top Countries by Total Medals

- Gold, Silver & Bronze Comparison
- Olympic Medal Trends by Year
- Athlete Gender Analysis
- Athlete Participation by Sport
- Average Medal Efficiency by Region
- Average Medal Count by Investment Category
- Medal Performance by Host Country

Summary of Results

The workbook provides a consolidated view of Olympic medal and athlete data. The visualizations allow users to compare medal totals and types, explore year-wise trends, review athlete participation, and examine differences across regions, host-country categories, and investment categories. Specific numerical findings should be added only after confirming the values displayed in the actual charts.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Transforms multiple CSV files into an accessible visual analysis.
- Supports comparisons across countries, years, medal types, sports, and regions.
- Interactive filters allow focused exploration where configured.
- Three dashboards and a Story organize results for presentation.
- Tableau provides a visual interface without requiring end users to write SQL queries.

Disadvantages / Limitations

- The analysis depends on the completeness and consistency of the supplied datasets.
- Comparisons are descriptive and do not establish causal relationships.
- No Snowflake warehouse or SQL-based transformation pipeline is part of the current implementation.
- The dashboards do not predict future Olympic results.
- A separate web application or automated refresh pipeline would require additional work.

9. CONCLUSION

Global Olympic Sports Performance Analysis demonstrates how Tableau can connect and explore Olympic CSV datasets through structured visualizations. The workbook includes eight worksheets, three dashboards, a four-point Story, configured dataset relationships, interactive filters where supported, and the Unique Athletes calculated field. It provides a visual basis for comparing medal performance and athlete participation while keeping conclusions grounded in the displayed data.

10. FUTURE SCOPE

- Develop a Flask web application to present or embed the published Tableau dashboards, if pursued.
- Add detailed data-quality checks and documented handling of missing or duplicate values.
- Add comparisons across Olympic editions, sports, and country groups where the data supports them.
- Introduce automated data refresh and repeatable preprocessing workflows.
- Explore predictive analytics only after defining a suitable target and validating data quality and model performance.

11. APPENDIX

11.1 Source Code (if any)

The current analysis was built in Tableau. No SQL scripts were used in the confirmed implementation. Include Tableau workbook files, calculated field details, or additional scripts in the repository if applicable.

11.2 Dataset Link

Institution-provided CSV datasets — Google Drive folder

11.3 GitHub & Project Demo Link

GitHub Repository

https://github.com/Aartigurjar310/Global-Olympic-Sports-Performance-Analysis

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End of Report